Good Data, Bad Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Good Data, Bad Data before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Recommendations

Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Confidence Is Not Correctness

Without opening the old lesson, explain one key idea from Confidence Is Not **Correctness**. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Save New Examples for Testing

Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Good Data, Bad Data trace and record the first step you can explain without help.

Use W2: Write a traceable **correction** for Data quality and documented repair. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Good Data, Bad Data explanation against the supplied facts before discussing it.

Use the worked case for Data quality and documented repair. Proposed explanation: “Every valid value is true”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Audit an independent paper-plane log

Each flight ID must be unique; distance is a whole number from 0 through 20 cm. Raw rows: F1 8 cm; F2 120 cm; F3 blank; F4 0 cm; another F1 8 cm. Source S1 confirms one F1 flight. S2 says F2 travelled 12 cm. S3 is blank.

S4 confirms zero. Preserve the raw rows, describe supported repairs and give the number of distinct flights. Report how many assessed flights travelled at least 10 cm, including missing records separately.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Good Data, Bad Data target to a trace and a justified conclusion.

> A table lists leaf lengths 8 cm, 90 mm, blank and 700 cm. Propose a documented cleaning action for each row. Which values can you safely change immediately?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Good Data, Bad Data case and show the deciding evidence.

Ordinary pencil lengths are recorded as 12 cm, 14 cm, blank and 130 cm.

> What needs checking? May 130 silently become 13?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A table labels a cat photo dog and lists the same photo three times. Identify both data problems and one sensible repair for each.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 21 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Good Data, Bad Data adds to Helpful Classroom Sorter and keep the evidence.

Audit the sorter's example cards and add a reason for including each field and condition.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A table lists leaf lengths 8 cm, 90 mm, blank and 700 cm. Propose a documented cleaning action for each row. Which values can you safely change immediately?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Check a schema without guessing

> Check a schema without guessing. Apply the supplied Good Data, Bad Data facts and keep your reasoning.

Use the RampLog **schema**. State the issue or valid status for height 60, travel blank and travel 0. Give the range allowed for each numeric field and explain why a blank and zero require different treatment.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Write a traceable correction

> Write a traceable correction. Apply the supplied Good Data, Bad Data facts and keep your reasoning.

C2 says T2 height is 6 cm and travel is 5 cm; the export says height 60 cm. Write a change-log entry with ID, field, old and new values with units, **evidence**, reason, reviewer and date. Use a fictional reviewer and date.

Explain where the raw export is kept.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Distinguish duplicate rows from similar trials

> Distinguish duplicate rows from similar trials. Apply the supplied Good Data, Bad Data facts and keep your reasoning.

The export repeats T1 with the same source C1, which confirms one trial. A different trial T5 also has height 4 and travel 3, with its own source C5 confirming a separate trial.

Explain which rows count once and why T5 must remain separate.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Audit an independent paper-plane log

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Audit an independent paper-plane log. Apply the supplied Good Data, Bad Data facts and keep your reasoning.

Each flight ID must be unique; distance is a whole number from 0 through 20 cm. Raw rows: F1 8 cm; F2 120 cm; F3 blank; F4 0 cm; another F1 8 cm. Source S1 confirms one F1 flight. S2 says F2 travelled 12 cm. S3 is blank.

S4 confirms zero. Preserve the raw rows, describe supported repairs and give the number of distinct flights. Report how many assessed flights travelled at least 10 cm, including missing records separately.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Handle conflicting and converted evidence

> Handle conflicting and converted evidence. Apply the supplied Good Data, Bad Data facts and keep your reasoning.

Two signed cards disagree about the same F3 flight: one says 9 cm and the other 11 cm. Neither is established as the **correction**. What can you report for the at-least-10-cm question?

Separately, a source for a new F5 flight explicitly records 100 mm. Convert it to centimetres and state the conversion **evidence** to retain.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Good Data, Bad Data

> A fictional toy-car distance table needs cleaning before use. Explain the labelled step before checking the changed condition.

A fictional toy-car distance table needs cleaning before use.

Contract: unique run ID; whole distance 0–30 cm. Raw rows: A=0 cm, B=200 mm, C=blank, another A=0 cm.

Source log confirms one A run and B's original measurement of 200 mm; C has no measurement.

AI ACADEMY / AI-1/21

Good Data, Bad Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Validate rows

A's zero is valid; B has wrong unit; C is missing; A is duplicated.

2. Evidence repair

200 mm / 10 =20 cm; source log supports retaining one A.

3. Keep unknown

C stays unresolved; do not replace it with 0.

4. Clean record

Usable unique rows: A=0 cm, B=20 cm; retain a separate change log.

CHANGE ONE THING

The source log gives B only as bare 200 with no unit.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which repairs are **evidence**-backed in the original case, and which row must remain unknown?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Keep raw sorter **input**s, validated records and an **evidence**-linked repair log as different artifacts.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Keep raw sorter **input**s, validated records and an **evidence**-linked repair log as different artifacts.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional toy-car distance table needs cleaning before use.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**.

1. Clean record

2. **Evidence** repair

3. Keep unknown

4. Validate rows

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Clean record Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**. Cleaning is not permission to make inconvenient values plausible.

The source log supports a unit conversion and removal of a **duplicate** copy. Zero means a measured zero distance; blank means no supplied measurement.

Keep the raw table and explain every change so someone can reconstruct what happened.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The source log gives B only as bare 200 with no unit. Which conclusion is supported by the supplied facts?

1. B remains unresolved rather than becoming 20 cm.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: B remains unresolved rather than becoming 20 cm. Without a confirmed unit, division by 10 would be an unsupported assumption.

Repair a missing building block

Review Recommendations (ai-1/20). Goal: Apply eligibility requirements before ranking candidates by preferences. Short practice: Why does the recommendation change without changing anyone’s preferences?.

Return to this week and explain what you changed.

Review Confidence Is Not **Correctness** (ai-1/18). Goal: Distinguish a confidence claim from **evidence** that a **prediction** was correct. Short practice: Was the unknown label guaranteed correct because its confidence was **80%**?.

Return to this week and explain what you changed.

Review Save New Examples for Testing (ai-1/15). Goal: Separate examples used to choose a model from examples reserved for its final evaluation. Short practice: Does 4/4 on the replay replace the original 3/4 result? Explain..

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

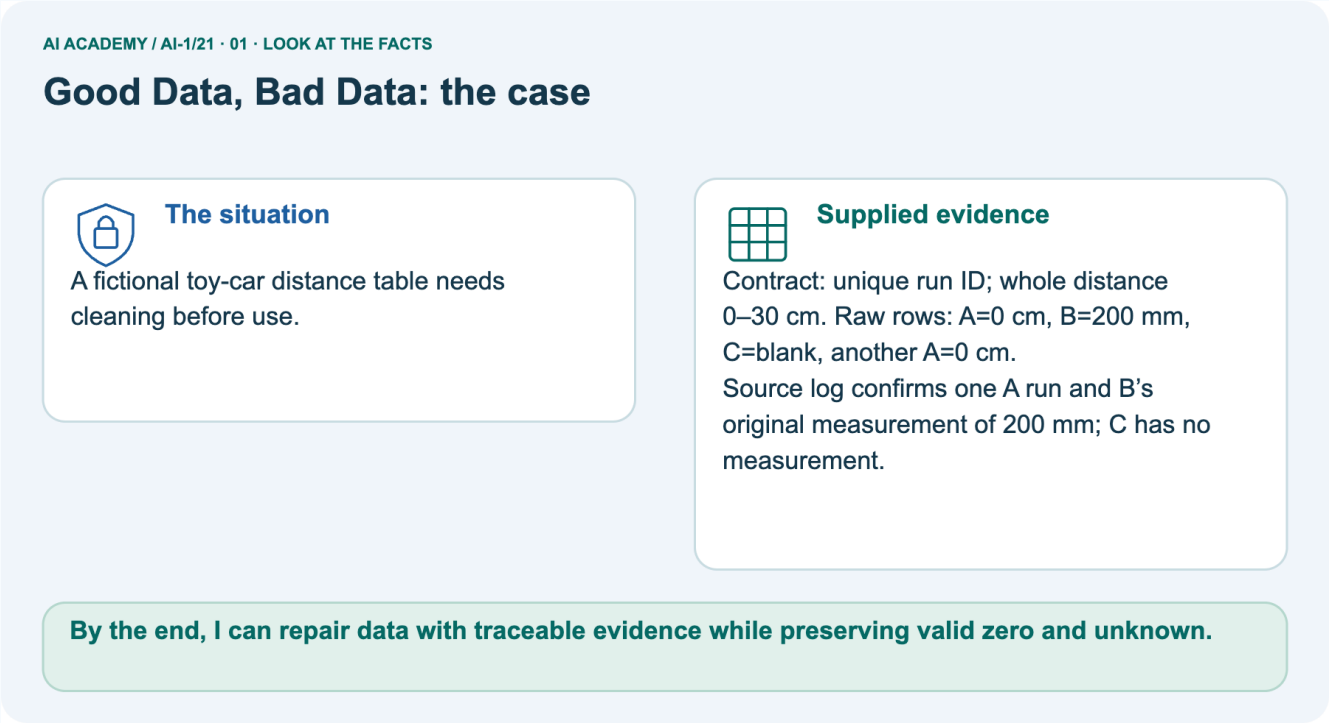
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Good Data, Bad Data picture from its labels, then record a relevant limit.

01. Good Data, Bad Data: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

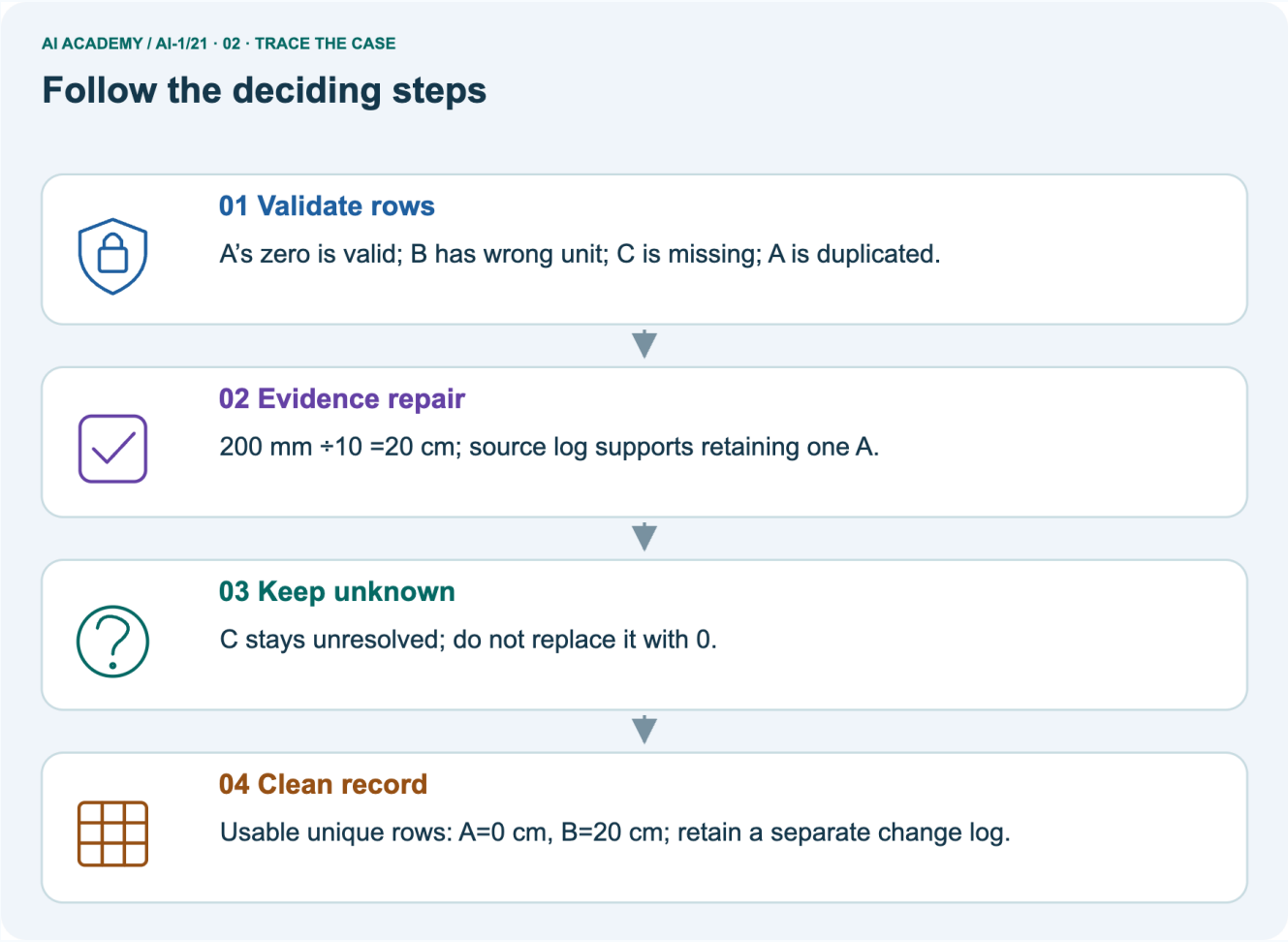
A fictional toy-car distance table needs cleaning before use. Supplied facts: Contract: unique run ID; whole distance 0–30 cm. Raw rows: A=0 cm, B=200 mm, C=blank, another A=0 cm.

Source log confirms one A run and B's original measurement of 200 mm; C has no measurement.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



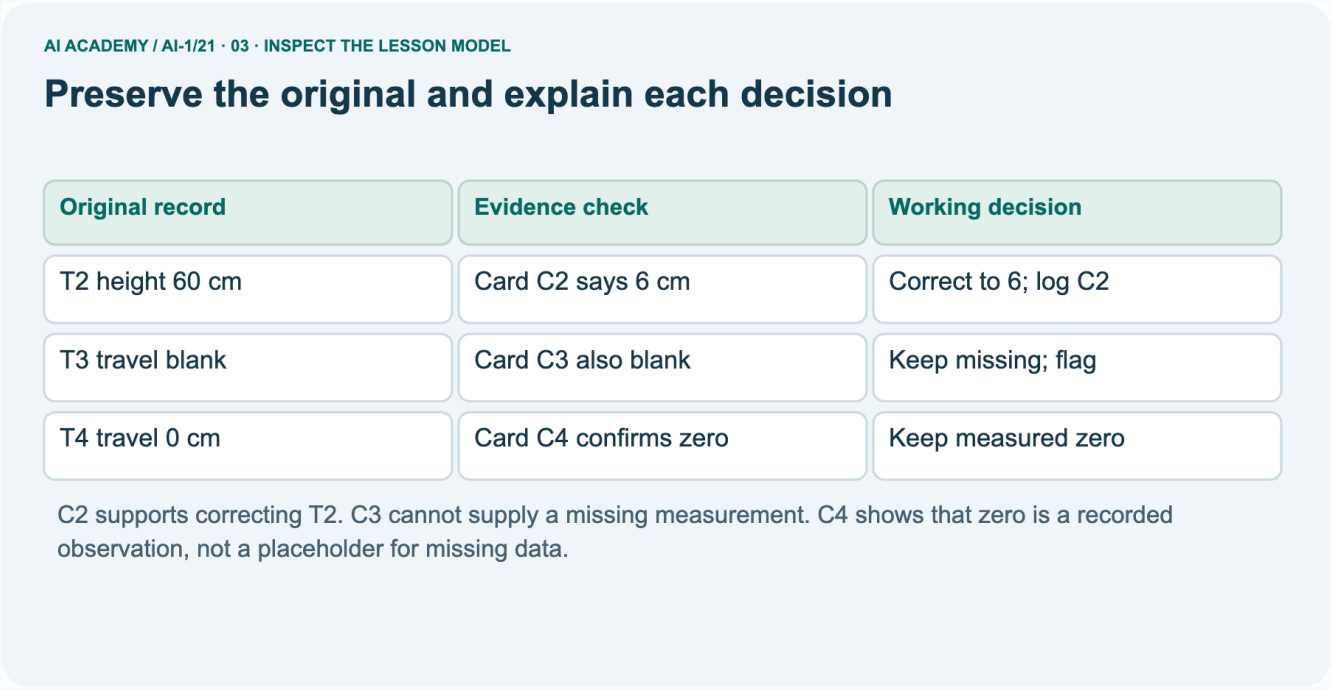
Follow the labelled picture with the matching explanation. Open the vector diagram

Validate rows: A's zero is valid; B has wrong unit; C is missing; A is **duplicate**d. → **Evidence** repair: 200 mm ÷10 =20 cm; source log supports retaining one A. → Keep unknown: C stays unresolved; do not replace it with 0. → Clean record: Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Preserve the original and explain each decision



Follow the labelled picture with the matching explanation. Open the vector diagram

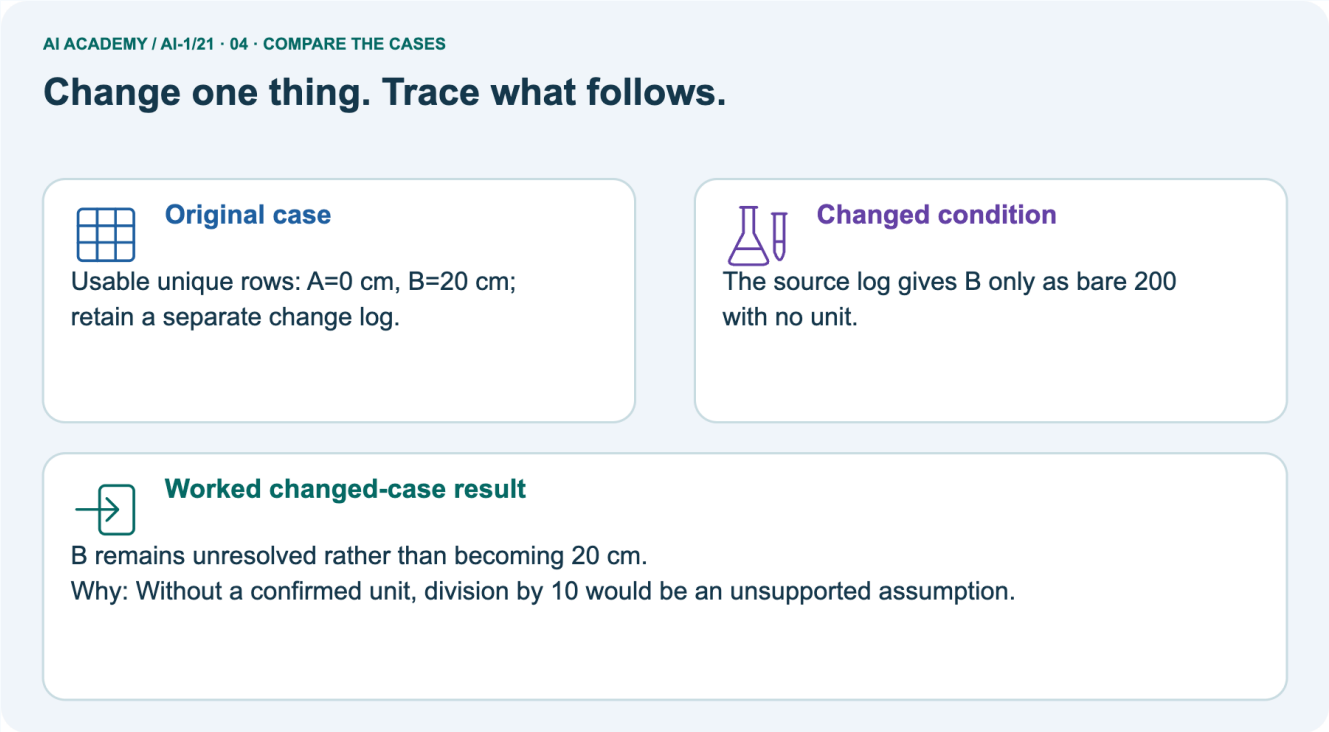
Preserve the original and explain each decision. T2 height 60 cm | Card C2 says 6 cm | Correct to 6; log C2. T3 travel blank | Card C3 also blank | Keep missing; flag.

T4 travel 0 cm | Card C4 confirms zero | Keep measured zero

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

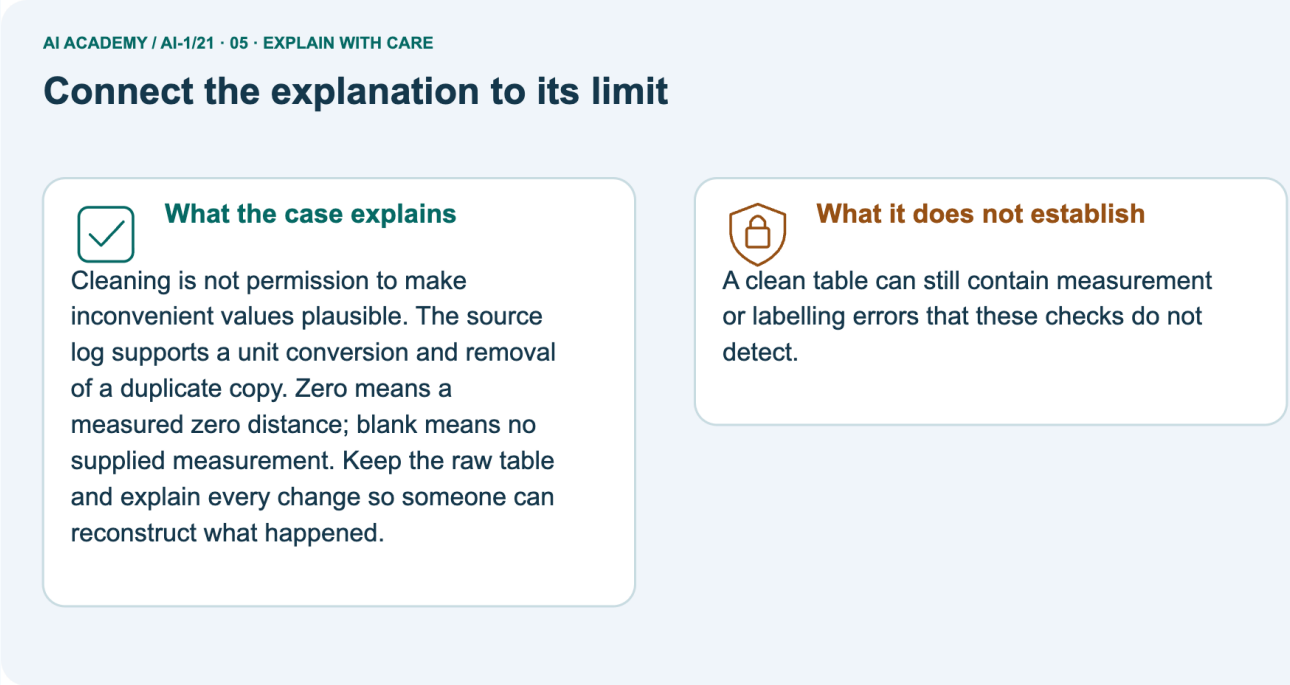
Original case: Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**. Changed condition: The source log gives B only as bare 200 with no unit. Expected result: B remains unresolved rather than becoming 20 cm.

Reason: Without a confirmed unit, division by 10 would be an unsupported assumption.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

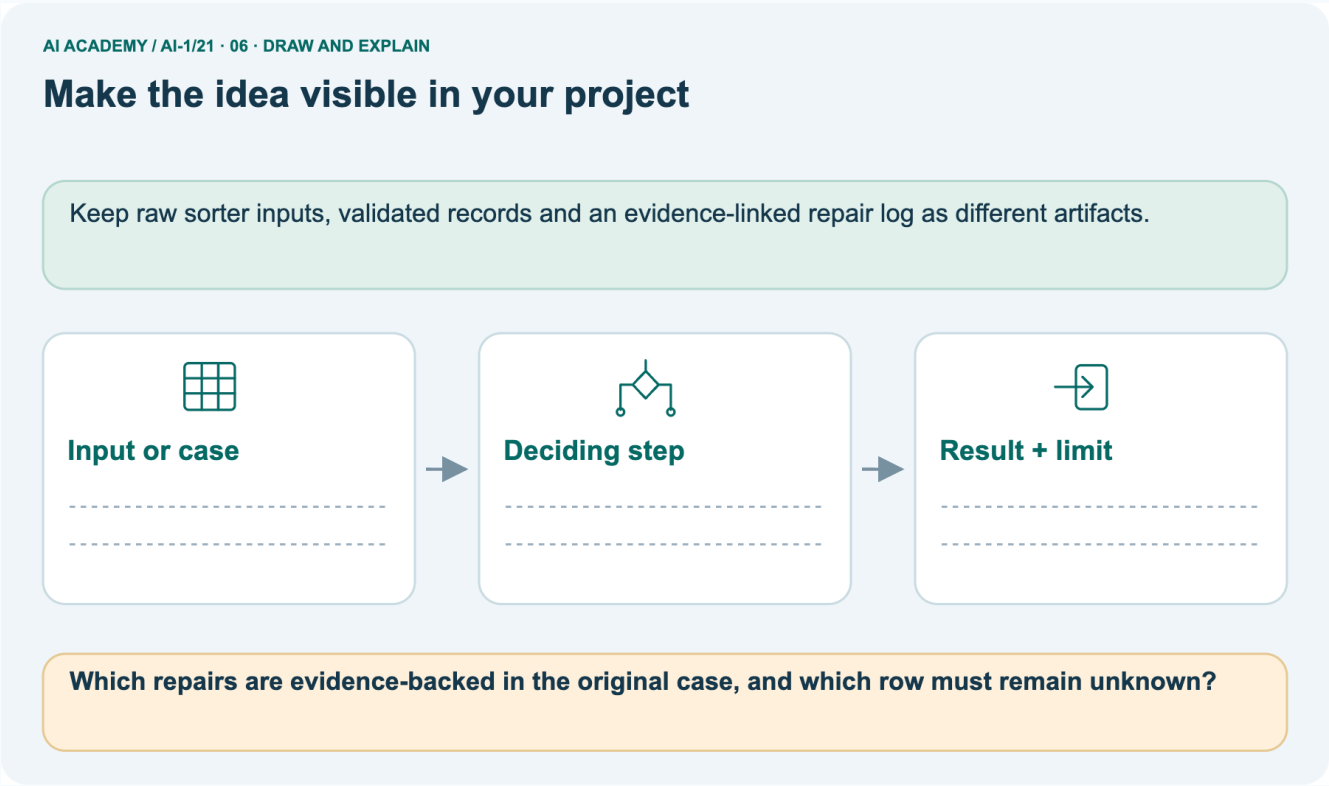
Cleaning is not permission to make inconvenient values plausible. The source log supports a unit conversion and removal of a **duplicate** copy. Zero means a measured zero distance; blank means no supplied measurement.

Keep the raw table and explain every change so someone can reconstruct what happened. Limit: A clean table can still contain measurement or labelling errors that these checks do not detect.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep raw sorter **input**s, validated records and an **evidence**-linked repair log as different artifacts. Draw your **input**, deciding step, result and limit.

Supported question: Which repairs are **evidence**-backed in the original case, and which row must remain unknown?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.